

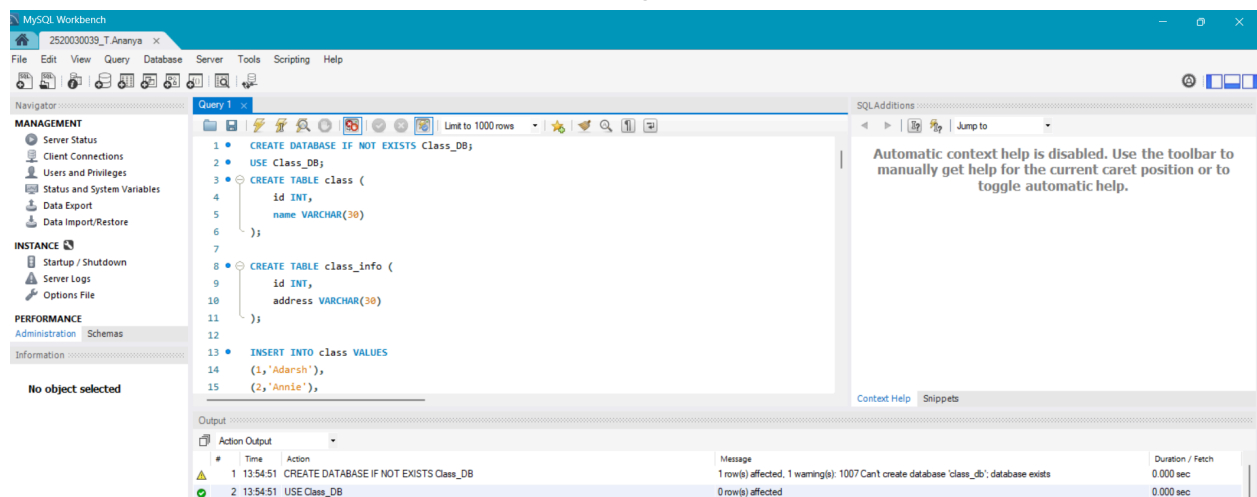
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

JOINS AND SET OPERATIONS

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Query-1:



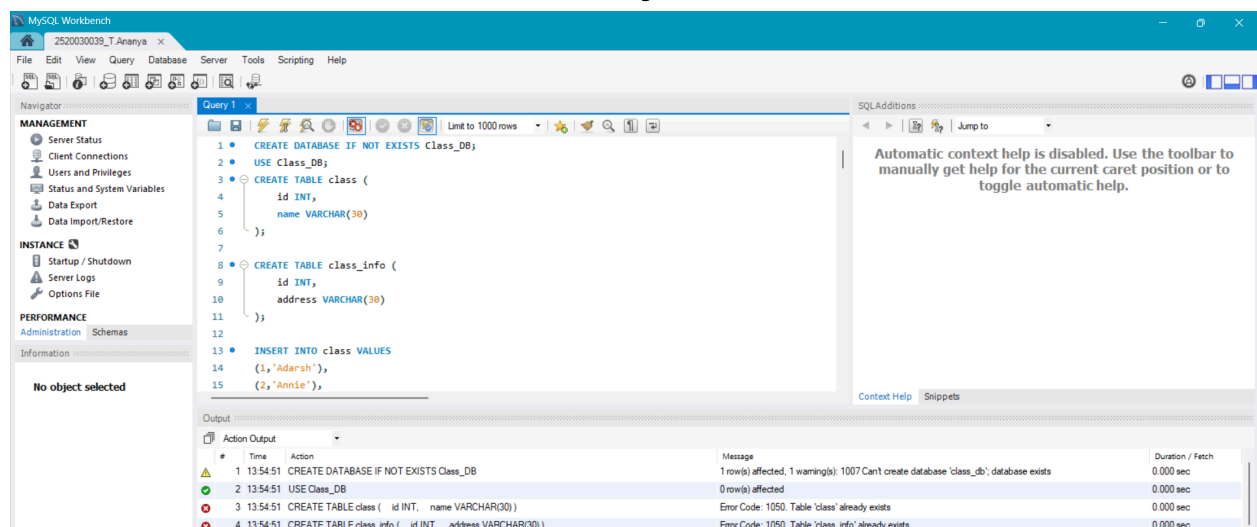
MySQL Workbench interface showing Query 1 execution. The query is as follows:

```
1 • CREATE DATABASE IF NOT EXISTS Class_DB;
2 • USE Class_DB;
3 • CREATE TABLE class (
4     id INT,
5     name VARCHAR(30)
6 );
7
8 • CREATE TABLE class_info (
9     id INT,
10    address VARCHAR(30)
11 );
12
13 • INSERT INTO class VALUES
14     (1,'Adarsh'),
15     (2,'Annie');
```

The Output pane shows the following results:

#	Time	Action	Message	Duration / Fetch
1	13:54:51	CREATE DATABASE IF NOT EXISTS Class_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'class_db'; database exists	0.000 sec
2	13:54:51	USE Class_DB	0 row(s) affected	0.000 sec

Query-2:



MySQL Workbench interface showing Query 2 execution. The query is identical to Query 1. The Output pane shows the following results:

#	Time	Action	Message	Duration / Fetch
1	13:54:51	CREATE DATABASE IF NOT EXISTS Class_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'class_db'; database exists	0.000 sec
2	13:54:51	USE Class_DB	0 row(s) affected	0.000 sec
3	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
4	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec

Query-3:

MySQL Workbench interface showing Query 3 execution. The query is as follows:

```
8 CREATE TABLE class_info (  
9     id INT,  
10     address VARCHAR(30)  
11 );  
12  
13 INSERT INTO class VALUES  
14     (1, 'Adarsh'),  
15     (2, 'Annie'),  
16     (4, 'Crew');  
17  
18 INSERT INTO class_info VALUES  
19     (1, 'DELHI'),  
20     (2, 'MUMBAI'),  
21     (3, 'Hyderabad');  
22
```

The output shows the following actions and messages:

#	Time	Action	Message
1	13:54:51	CREATE DATABASE IF NOT EXISTS Class_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database
2	13:54:51	USE Class_DB	0 row(s) affected
3	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists
4	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists
5	13:54:51	INSERT INTO class VALUES (1,'Adarsh'), (2,'Annie'), (4,'Crew')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
6	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'Hyderabad')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Query-4:

MySQL Workbench interface showing Query 4 execution. The query is as follows:

```
21 (3, 'Hyderabad');  
22  
23 SELECT *  
24 FROM class  
25 CROSS JOIN class_info;  
26  
27 CREATE TABLE class (  
28     id INT,  
29     name VARCHAR(30)  
30 );  
31  
32 CREATE TABLE class_info (  
33     id INT,  
34     address VARCHAR(30)  
35 );
```

The output shows the following actions and messages:

#	Time	Action	Message	Duration / Fetch
1	13:54:51	CREATE DATABASE IF NOT EXISTS Class_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'class_db'; database exists	0.000 sec
2	13:54:51	USE Class_DB	0 row(s) affected	0.000 sec
3	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
4	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec
5	13:54:51	INSERT INTO class VALUES (1,'Adarsh'), (2,'Annie'), (4,'Crew')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
6	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'Hyderabad')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
7	13:54:51	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	361 row(s) returned	0.000 sec / 0.000 sec
8	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
9	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec

Query-5:

MySQL Workbench interface showing Query 1. The SQL script defines tables and inserts data:

```
33 id INT,  
34 address VARCHAR(30)  
35 );  
36  
37 INSERT INTO class VALUES  
38 (1,'Adarsh'),  
39 (2,'Annie'),  
40 (3,'Crew'),  
41 (4,'Annu');  
42  
43 INSERT INTO class_info VALUES  
44 (1,'DELHI'),  
45 (2,'MUMBAI'),  
46 (3,'CHENNAI');  
47
```

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	13:54:51	CREATE DATABASE IF NOT EXISTS Class_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'class_db'; database exists	0.000 sec
2	13:54:51	USE Class_DB	0 row(s) affected	0.000 sec
3	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
4	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec
5	13:54:51	INSERT INTO class VALUES (1,'Adarsh'), (2,'Annie'), (4,'Crew')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
6	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'Hyderabad')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
7	13:54:51	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	361 row(s) returned	0.000 sec / 0.000 sec
8	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
9	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec
10	13:54:51	INSERT INTO class VALUES (1,'Adarsh'), (2,'Annie'), (3,'Crew'), (4,'Annu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
11	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Query-6:

MySQL Workbench interface showing Query 1. The SQL script continues from Query 5:

```
46 (3,'Hyderabad');  
47  
48 SELECT *  
49 FROM class  
50 INNER JOIN class_info  
51 ON class.id = class_info.id;  
52  
53 SELECT class.name,  
54 class_info.address  
55 FROM class  
56 INNER JOIN class_info  
57 ON class.id=class_info.id;  
58  
59 SELECT *  
60 FROM class
```

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
4	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec
5	13:54:51	INSERT INTO class VALUES (1,'Adarsh'), (2,'Annie'), (4,'Crew')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
6	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'Hyderabad')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
7	13:54:51	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	361 row(s) returned	0.000 sec / 0.000 sec
8	13:54:51	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
9	13:54:51	CREATE TABLE class_info (id INT, address VARCHAR(30))	Error Code: 1050. Table 'class_info' already exists	0.000 sec
10	13:54:51	INSERT INTO class VALUES (1,'Adarsh'), (2,'Annie'), (3,'Crew'), (4,'Annu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
11	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
12	13:54:51	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	90 row(s) returned	0.000 sec / 0.000 sec
13	13:54:51	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id=class_info.id	90 row(s) returned	0.000 sec / 0.000 sec

Query-7:

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Query 1

```

59 SELECT *
60 FROM class
61 NATURAL JOIN class_info;
62
63 INSERT INTO class VALUES
64 (5,'ashish');
65
66 INSERT INTO class_info VALUES
67 (7,'NOIDA'),
68 (8,'PANIPAT');
69
70 SELECT *
71 FROM class
72 LEFT OUTER JOIN class_info
73 ON class.id=class_info.id;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
11	13:54:51	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
12	13:54:51	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	90 row(s) returned	0.000 sec / 0.000 sec
13	13:54:51	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id=class_info.id	90 row(s) returned	0.000 sec / 0.000 sec
14	13:54:51	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	90 row(s) returned	0.015 sec / 0.000 sec
15	13:54:51	INSERT INTO class VALUES (5,'ashish')	1 row(s) affected	0.000 sec
16	13:54:51	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT');	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec

Query-8:

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

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INSTANCE

- Startup / Shutdown
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- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Query 1

```

68 (8,'PANIPAT');
69
70 SELECT *
71 FROM class
72 LEFT OUTER JOIN class_info
73 ON class.id=class_info.id;
74
75 SELECT *
76 FROM class
77 LEFT JOIN class_info
78 ON class.id=class_info.id
79 WHERE class_info.id IS NULL;
80
81 SELECT *
82 FROM class

```

SQLAdditions

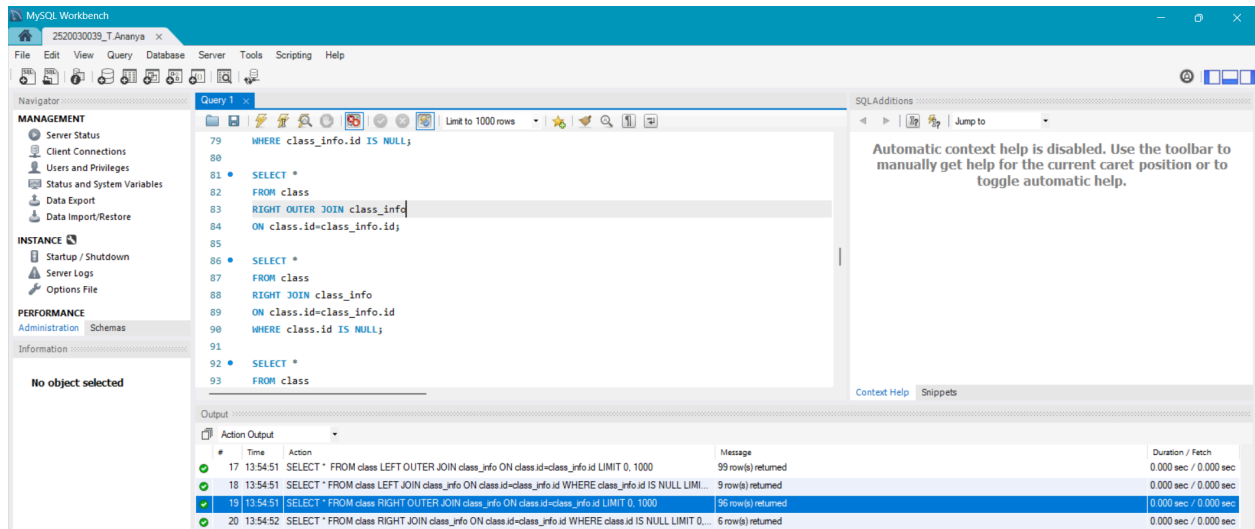
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

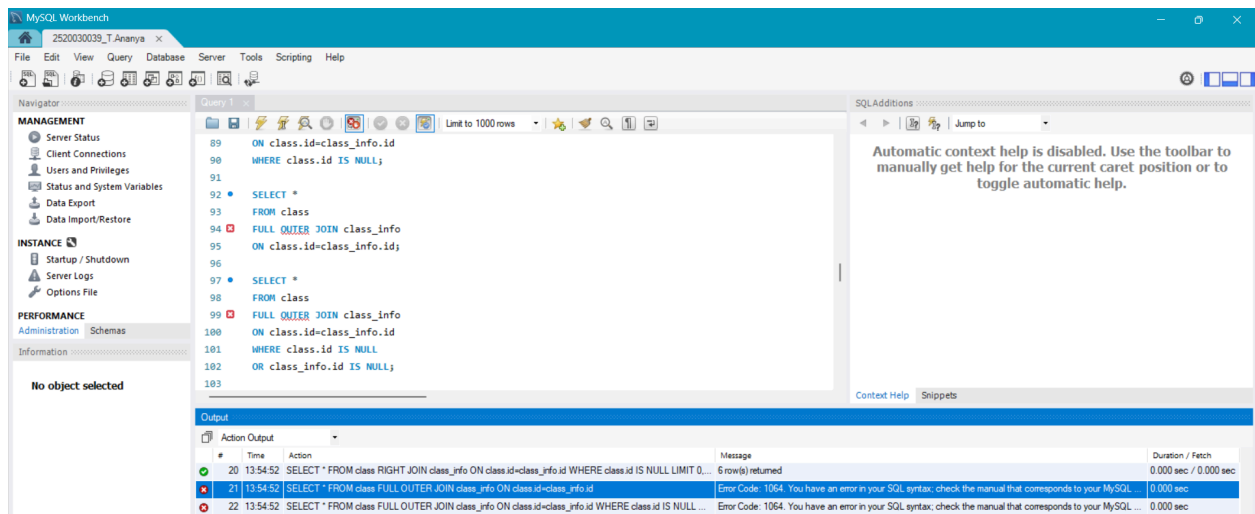
Output

#	Time	Action	Message	Duration / Fetch
17	13:54:51	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 1000	99 row(s) returned	0.000 sec / 0.000 sec
18	13:54:51	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id IS NULL LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
19	13:54:51	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 1000	96 row(s) returned	0.000 sec / 0.000 sec

Query-9:



Query-10:



Query-11:

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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INSTANCE

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- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Query 1

```

104 CREATE TABLE first_table(
105   id INT,
106   name VARCHAR(30)
107 );
108
109 CREATE TABLE second_table(
110   id INT,
111   name VARCHAR(30)
112 );
113
114 INSERT INTO first_table VALUES
115   (1,'Adarsh'),
116   (2,'Annie');
117
118 INSERT INTO second_table VALUES

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

#	Time	Action	Message	Duration / Fetch
20	13:54:52	SELECT * FROM class RIGHT JOIN class_info ON class id=class_info id WHERE class id IS NULL LIMIT 0...	6 row(s) returned	0.000 sec / 0.000 sec
21	13:54:52	SELECT * FROM class FULL OUTER JOIN class_info ON class id=class_info id	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec
22	13:54:52	SELECT * FROM class FULL OUTER JOIN class_info ON class id=class_info id WHERE class id IS NULL ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec
23	13:54:52	CREATE TABLE first_table(id INT, name VARCHAR(30))	Error Code: 1050. Table 'first_table' already exists	0.000 sec
24	13:54:52	CREATE TABLE second_table(id INT, name VARCHAR(30))	Error Code: 1050. Table 'second_table' already exists	0.000 sec
25	13:54:52	INSERT INTO first_table VALUES (1,'Adarsh'), (2,'Annie')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec

Query-12:

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
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INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Query 1

```

116 (2,'Annie');
117
118 INSERT INTO second_table VALUES
119   (2,'Annie'),
120   (3,'appolamma');
121
122 SELECT * FROM first_table
123 UNION
124 SELECT * FROM second_table;
125
126 SELECT name FROM first_table
127 UNION
128 SELECT name FROM second_table;
129
130 SELECT * FROM first_table

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

#	Time	Action	Message	Duration / Fetch
20	13:54:52	SELECT * FROM class RIGHT JOIN class_info ON class id=class_info id WHERE class id IS NULL LIMIT 0...	6 row(s) returned	0.000 sec / 0.000 sec
21	13:54:52	SELECT * FROM class FULL OUTER JOIN class_info ON class id=class_info id	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec
22	13:54:52	SELECT * FROM class FULL OUTER JOIN class_info ON class id=class_info id WHERE class id IS NULL ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec
23	13:54:52	CREATE TABLE first_table(id INT, name VARCHAR(30))	Error Code: 1050. Table 'first_table' already exists	0.000 sec
24	13:54:52	CREATE TABLE second_table(id INT, name VARCHAR(30))	Error Code: 1050. Table 'second_table' already exists	0.000 sec
25	13:54:52	INSERT INTO first_table VALUES (1,'Adarsh'), (2,'Annie')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
26	13:54:52	INSERT INTO second_table VALUES (2,'Annie'), (3,'appolamma')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec

Query-13:

MySQL Workbench

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Navigator

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- Server Status
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INSTANCE

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PERFORMANCE

- Administration
- Schemas

Information

No object selected

Query 1

```

128 SELECT name FROM second_table;
129
130 SELECT * FROM first_table
131 UNION ALL
132 SELECT * FROM second_table;
133
134 SELECT COUNT(*)
135 FROM
136 (
137 SELECT * FROM first_table
138 UNION ALL
139 SELECT * FROM second_table
140 ) As;
141
142 SELECT * FROM first_table

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
26	13:54:52	INSERT INTO second_table VALUES (2,'Annie'), (3,'appalamma')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
27	13:54:52	SELECT * FROM first_table UNION SELECT * FROM second_table	6 row(s) returned	0.000 sec / 0.000 sec
28	13:54:52	SELECT name FROM first_table UNION SELECT name FROM second_table	6 row(s) returned	0.000 sec / 0.000 sec
29	13:54:52	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	12 row(s) returned	0.000 sec / 0.015 sec
30	13:54:52	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) A...	1 row(s) returned	0.000 sec / 0.000 sec

Query -14:

MySQL Workbench

2520030039_T_Ananya

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

Information

No object selected

Query 1

```

140 ) As;
141
142 SELECT * FROM first_table
143 INTERSECT
144 SELECT * FROM second_table;
145
146 SELECT name FROM first_table
147 INTERSECT
148 SELECT name FROM second_table;
149
150 SELECT * FROM first_table
151 MINUS
152 SELECT * FROM second_table;
153
154

```

SQLAdditions

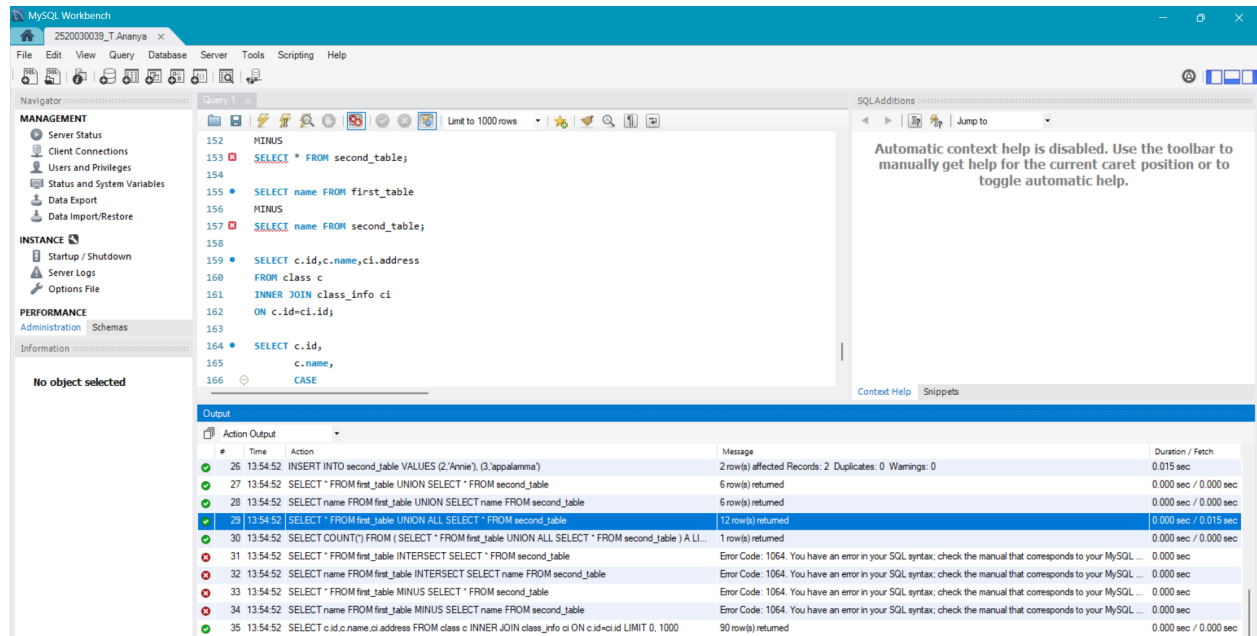
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

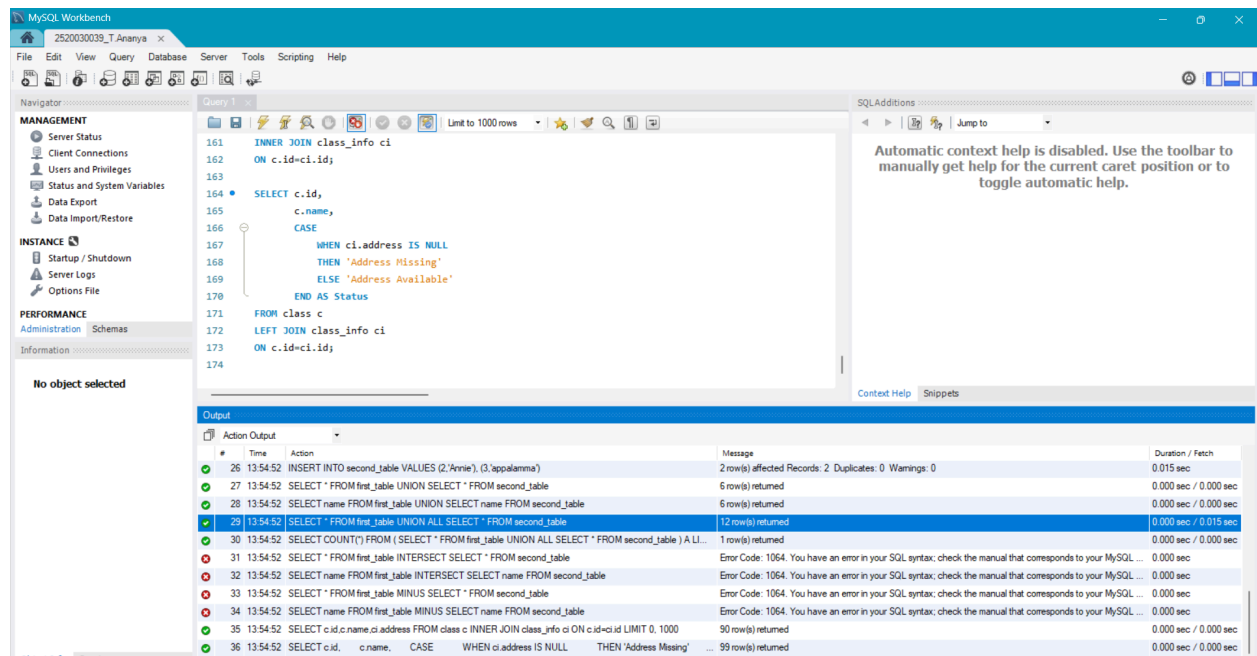
Output

#	Time	Action	Message	Duration / Fetch
26	13:54:52	INSERT INTO second_table VALUES (2,'Annie'), (3,'appalamma')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.015 sec
27	13:54:52	SELECT * FROM first_table UNION SELECT * FROM second_table	6 row(s) returned	0.000 sec / 0.000 sec
28	13:54:52	SELECT name FROM first_table UNION SELECT name FROM second_table	6 row(s) returned	0.000 sec / 0.000 sec
29	13:54:52	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	12 row(s) returned	0.000 sec / 0.015 sec
30	13:54:52	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) A LI...	1 row(s) returned	0.000 sec / 0.000 sec
31	13:54:52	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec
32	13:54:52	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec
33	13:54:52	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ...	0.000 sec

Query -15:



Query-16:



Output:

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	id	name			
	1	abhi			
	2	adam			
	1	Adarsh			
	2	Annie			
	3	chester			
▶	3	appalamma			

Result 19 Result 20 Result 21 Result 22 **Result 23** × Result 24 Result 25 Result 26 Result 27 Result 28

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Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	id	name	id	address			
▶	1	abhi	1	DELHI			
	1	abhi	1	DELHI			
	1	abhi	1	DELHI			
	1	abhi	1	DELHI			
	1	Adarsh	1	DELHI			
	1	Adarsh	1	DELHI			
	1	Adarsh	1	DELHI			
	1	Adarsh	1	DELHI			

Result 19 Result 20 **Result 21** × Result 22 Result 23 Result 24 Result 25 Result 26 Result 27 Result 28